

HAT-P-27b

R-0048

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-27_260217 · created 2026-07-09T05:50:45+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461085.831 +/- 0.0088 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.039
Transit depth (Rp/Rs)^2	1.92 +/- 1.08 [%]
Orbital Inclination (inc)	88.62 +/- 2.93
Airmass coefficient 1 (a1)	1.77 +/- 1.06
Airmass coefficient 2 (a2)	-0.333 +/- 0.064
Scatter in the residuals of the lightcurve fit is	47.37 %
Best Comparison Star	#2 - [339, 341]
Optimal Aperture	1.93
Optimal Annulus	4.75
Transit Duration (day)	0.047 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.138 $\pm$ 0.039 vs 0.119 (deviation 0.49 σ)
Duration fit vs published	0.047 d vs 0.07 d (deviation 2.3 σ)
Mid-time O-C	-4391.0 min
Depth SNR	3.5
Residual scatter	47.37 %

Light curve

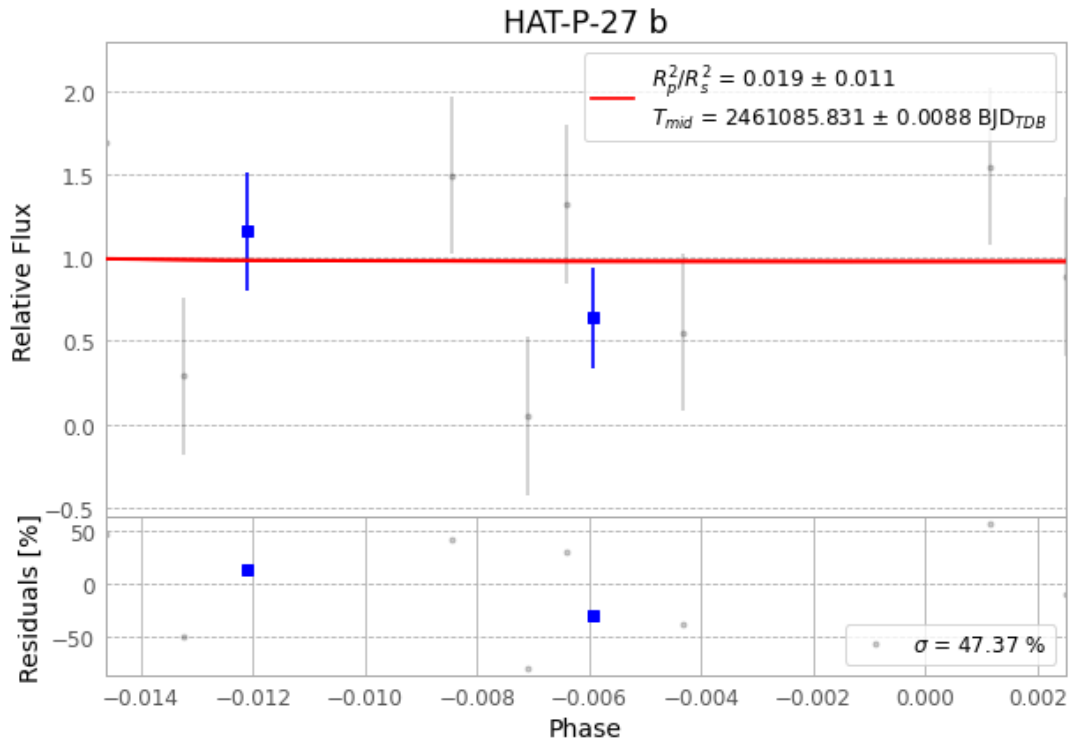


Figure 1 — FinalLightCurve_HAT-P-27 b_2026-02-17.png (SHA-256 67bc3fd76c0be7c6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [427, 125], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a1fff2d0fa231ea7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

39 FITS frames (26 MB), HATP-27260217072715.FITS → HATP-27260217092115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-27-260217.log (9dbd4a7baa55...), FinalLightCurve_HAT-P-27 b_2026-02-17.png (67bc3fd76c0b...), FinalLightCurve_HAT-P-27 b_2026-02-17.csv (a8d957eb1cfd...)

Compute resources

Wall time 97.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.